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PHY 102 — Past Questions & Materials

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Course: PHY 102

Institution: University of Port Harcourt

Papers in this pack: 1

Levels: 100

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Paper 1 of 1 — Introduction To heat and thermodynamic PHY 102

Level 100 • Second Semester • PastQuestion

UNIVERSITY OF ABUJA

FACULTY OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER 2016/2017 EXAMINATION

COURSE CODE: PHY 102

COURSE TITLE: INTRODUCTORY HEAT AND THERMODYNAMICS.

INSTRUCTION: ATTEMPT ALL QUESTIONS IN SECTION A AND
TWO QUESTIONS IN SECTION B.

TIME ALLOWED: 2 HRS

ALL SYMBOLS HAVE THEIR USUAL MEANINGS.

SECTION A

- a) Obtain the black body radiation curve.
- b) can travel through space and is the means by which heat reaches the earth from the sun.
- c) Obtain the thermodynamic temperature scale.
- d) Emissivity of radiation can be defined as
- e) What is thermal efficiency of a cycle?
- f) How different is Convection from Radiation in the transmission of heat.
- g) A steel ruler calibrated at 200C. what will be the error in measuring a distance of 25cm at 380C (α of steel = 1.2×10^{-5}).
- h) At what temp is the Fahrenheit and Celsius scale the same?
- i) Write five different forms of ideal gas equation.

SECTION B

- Q1 (a) Define thermal conductivity.
- (b) Calculate the steady temperature of a tungsten filament of 100w, if the filament length is 0.5m and radius 0.1mm, assume it to be a black body.
- Q2 (a) Define Cubic expansivity, γ .
- (b) A gas has a volume of 80m³ at STP. Calculate the new temperature and pressure if the gas is compressed adiabatically to a volume of 45cm³. ($C_p=1.6\text{kJ/kg-1K-1}$, $C_v=1.20\text{kJ/kg-1K-1}$).
- (c) State Clausius' statement of second law of thermodynamics.
- Q3 (a) Explain Pre'vost's theory of heat exchange.
- (b) Two thermometers, one graduated in Fahrenheit and the other in Celsius are placed in a bath and the bath is gradually heated. At what temperature will both thermometers read the same?
- Q4 (a) State Avogadro's law.
- (b) When the pressure of a fixed mass of gas is doubled at constant temp T, by what fraction does it volume change?
- (c) When is a system said to be isobaric?